

GREEN BUCKS

TO ACHIEVE ITS NET-ZERO EMISSIONS TARGET BY 2070, THE INDIAN ECONOMY WOULD REQUIRE INVESTMENTS WORTH MORE THAN \$10 TRILLION. BUT WHAT WILL THE RETURN ON THIS INVESTMENT LOOK LIKE?

BY NIDHI SINGAL

LOWERING CARBON emissions is serious business. Serious enough to draw financing of \$94.8 trillion over the next 38 years to meet the climate goals of just eight emerging economies—China, India, Indonesia, Kenya, South Africa, the UAE, Nigeria and South Africa—says a report by Standard Chartered. Of this, China needs \$35.1 trillion, while India's requirement is pegged at \$12.4 trillion till 2060. However, an independent study by the CEEW Centre for Energy Finance estimated that India will need investments of \$10.1 trillion by 2070, says Divya Datt, a Programme

Management Officer at the India Office of the UN Environment Programme. "Of this, \$8.4 trillion is required to scale up power generation, integration, distribution and transmission infrastructure. Another \$1.5 trillion is required for green hydrogen production capacity." While different reports point to varied estimates, the ballpark figure remains the same for the next 48 years.

A sector-wise break-up of the financial requirement, though, is only available for the next decade as industries, businesses and the government plan for 5-10 years, and not for 50. Nonetheless, consider-

ing carbon emissions of 2.44 billion tonnes in 2020, India has made big commitments, and appears to be on track to achieve them.

Sector-wise investments

The biggest emitters of greenhouse gases in India continue to be power generation units, industries—primarily steel and cement—and transport, and big investments are already deployed to tackle them. Planned investments for the next decade for non-fossil fuel industries are around ₹35-45 lakh crore. As part of this, the government has introduced the production-linked incentive (PLI)

PLANNED INVESTMENTS

POWER GENERATION

Non-fossil fuel:

₹35-45 lakh crore

Includes investments towards transmission infrastructure and energy storage capabilities

INDUSTRY

Steel and cement: ₹65,000 crore + Investments across private companies

TRANSPORT

CNG: ₹1.8-2.2 lakh crore

Includes 8th, 9th, 10th and 11th rounds of investment by the PNGRB

Ethanol: ₹28,000-32,000 crore Investments to be made by industry and the government through PSUs

SOURCE ICRA